

A small coil is positioned so that its axis lies along the axis of a large bar magnet, as shown in Fig. 6.1.

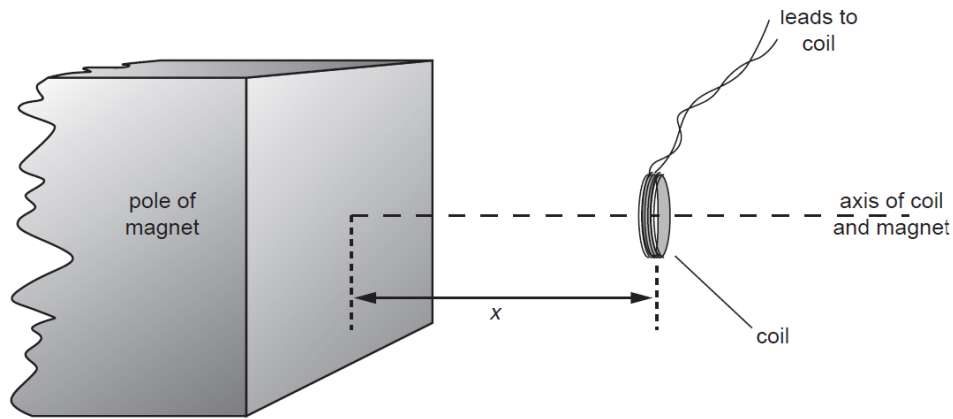


Fig. 6.1

The coil has a cross-sectional area of 0.40 cm^2 and contains 150 turns of wire.

The average magnetic flux density B through the coil varies with distance x between the face of the magnet and the plane of the coil as shown in Fig. 6.2.

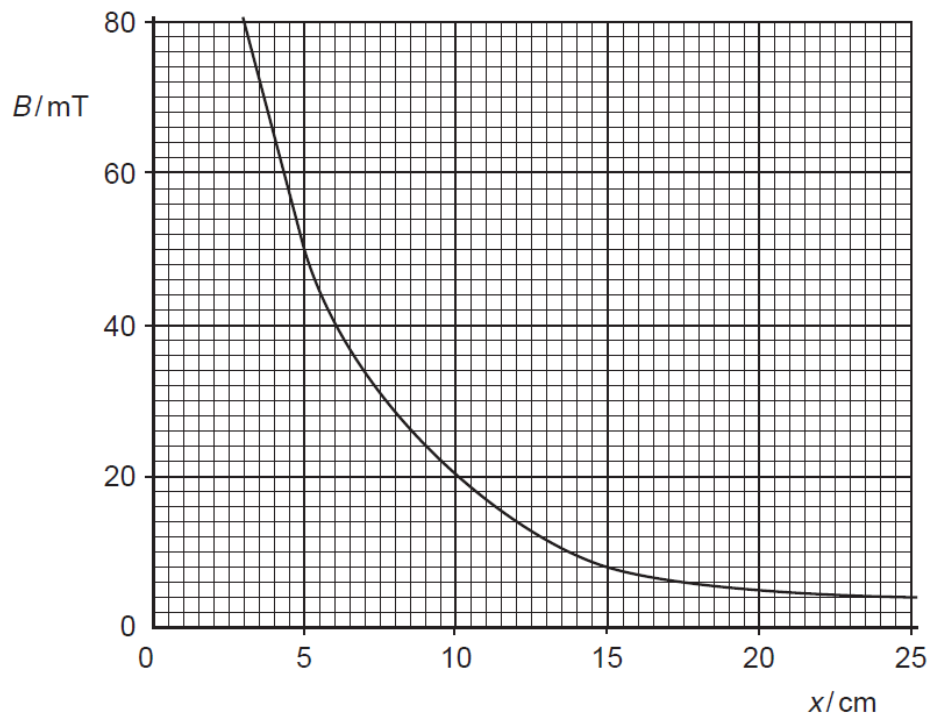


Fig. 6.2

- (a) (i) The coil is 5.0 cm from the face of the magnet. Use Fig. 6.2 to determine the magnetic flux density in the coil

magnetic flux density = T [1]

(ii) Hence show that the magnetic flux linkage of the coil is 3.0×10^{-4} Wb.

[2]

(b) The coil is moved along the axis of the magnet so that the distance x changes from $x = 5.0$ cm to $x = 15.0$ cm in a time of 0.30 s. Calculate

(i) the change in flux linkage of the coil,

change in flux linkage = Wb [1]

(ii) the average induced e.m.f. induced in the coil

e.m.f. = V [2]

(c) State and explain the variation, if any, of the speed of the coil so that the induced e.m.f. remains constant during the movement in (b).

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 [2]

(d) Use Lenz's law to explain why work has to be done to move the coil along the axis shown in Fig 6.1.

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 [2]

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[The Singapore Mass Rapid Transit (SMRT) started its first train services in 1997. It was a massive